TimeTraveler: Temporal Strategy Planning with Time Dictionary for Streaming Video Understanding

Minkuk Kim[1], Junho Kim[2†], Seong Tae Kim[1†]

[1]Kyung Hee University [2]University of Illinois Urbana-Champaign [†]Corresponding authors

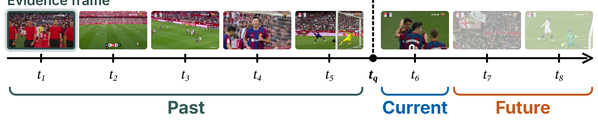
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TASK

Streaming Video QA

Which logos were visible as the players and referees entered the field?

Evidence frame



Over an **unbounded streaming horizon**, questions are **unpredictable**. Yet **storing every frame is infeasible** (**118.65 GB** KV cache per hour).

Memory-based Methods



Streamed frames

- Historical memory of agent
- Pruned tokens as memory
- Selective recall in KV memory

↓ *Only lookback*

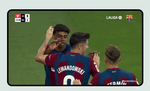


Detail memory preservation? ✗



Temporal planning? ✗

TimeTraveler (Ours)

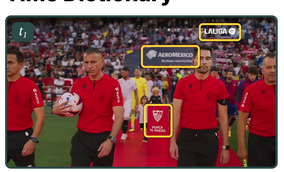


Context at t_q

Looks **past-related**. Let's **look back**. ✓

- **Past Lookback** ✓
- Current Perception
- Future Observation

Time Dictionary *entry d_i at t_i*



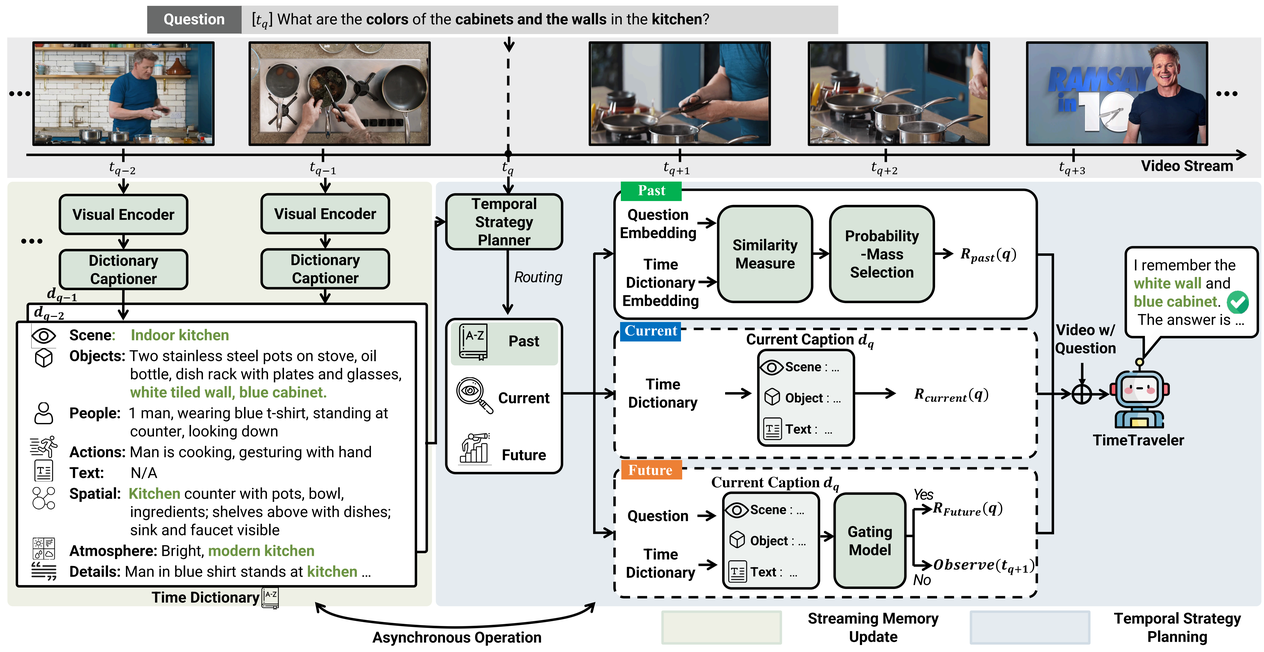
- Scene: Stadium
- People: Referees, players, crowd
- Actions: Player walkout
- Text: AeroMexico, LALIGA, NUN...

I found the logos in the **Time Dictionary**! The answer is ... ✓

TL;DR In streaming video QA, the **streaming horizon is unbounded**, and the model **cannot know when a question will come or what it will ask**. **TimeTraveler** keeps fine-grained details in a cheap text memory, a **Time Dictionary** of structured captions, and **plans which moment to read** (past, current, or future) based on each question's intent. Training-free, it improves Qwen3-VL-8B and InternVL3.5-8B on OVO-Bench and StreamingBench.

Project arXiv

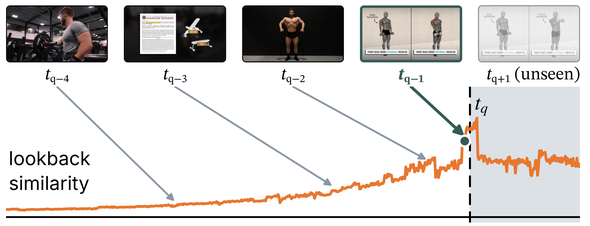
Time Dictionary + Temporal Strategy Planning



Qualitative Example: Past Lookback

The retrieved entry keeps the on-screen text and the red-highlighted muscles.

Question What muscle groups are highlighted in red on the left side of the screen during the front raise exercise?



Retrieved Time Dictionary entry at t_{q-1}

Scene	Indoor, textured background
Objects	Two 3D human figures with muscle anatomy diagrams, holding dumbbells
People	One figure on the left, one on the right ...
Actions	The figures are standing, performing front raises with dumbbells ...
Text	FRONT RAISE (MORE EXTERNALLY...
Spatial	The two figures are side-by-side, each in a different pose
Atmosphere	Neutral lighting, monochromatic color scheme with red highlights on deltoids
Details	...highlighting the triceps and chest muscles.

Main Results

- Best online method on both benchmarks: +6.32 (OVO-Bench) and +5.18 (StreamingBench) over Qwen3-VL-8B.
- With the same Qwen2.5-VL-7B backbone, it exceeds QueryStream by 5.01 and StreamAgent by 3.12 on OVO-Bench.

Results on OVO-Bench and StreamingBench

Shaded: TimeTraveler on each backbone, with the gain over that backbone.

		OVO-Bench				StreamingBench			
Model	**#Frames**	**Real-Time**	**Backward**	**Forward**	**Overall**	**Real-Time**	**Omni**	**Contextual**	**Overall**
Flash-VStream-7B	1 fps	28.37	27.38	45.09	33.61	23.23	26.00	24.12	24.04
Dispider-7B	1 fps	54.55	36.06	34.72	41.78	67.63	35.66	33.61	53.12
TimeChat-Online-7B	1 fps	58.60	42.00	36.40	45.60	75.36	–	–	–
StreamAgent-7B	1 fps	61.30	41.70	45.40	49.40	–	–	–	–
QueryStream-7B	1 fps	61.40	42.10	39.03	47.51	75.32	–	–	–
InternVL3.5-8B	1 fps	68.69	32.25	42.59	47.84	77.60	41.90	41.80	61.71
+ TimeTraveler	1 fps	75.30	34.72	37.06	49.03 +1.19	80.72	45.50	48.10	**65.64** +3.93
Qwen3-VL-8B	1 fps	63.24	43.44	48.42	51.70	72.80	43.80	36.75	59.33
+ TimeTraveler	1 fps	73.28	51.37	49.41	**58.02** +6.32	80.16	40.70	49.20	64.51 +5.18

Analysis

Effect of memory representation

OVO-Bench under a similar memory-token budget (Qwen3-VL-8B). RT: real-time; BW: backward; FW: forward.

Memory	RT	BW	FW	All
No memory	63.24	43.44	48.42	51.70
Visual prototype	66.19	44.16	51.36	53.90
Flat text	70.98	48.57	47.83	55.79
Time Dictionary (Ours)	**73.28**	**51.37**	**49.41**	**58.02**

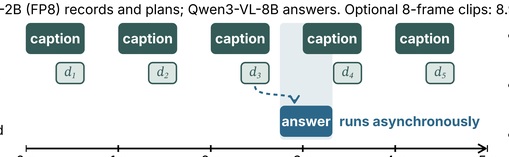
Effect of temporal routing

OVO-Bench Backward Tracing (past questions). EPM: episodic memory; HLD: hallucination detection; ASI: action sequence identification. †Mean of five runs.

Routing	EPM	HLD	ASI	Avg.
No memory	52.53	10.22	67.57	43.44
All-to-past lookback	57.91	25.81	67.57	50.43
Random routing†	52.12	19.78	63.24	45.05
Our routing	**59.26**	**25.27**	**69.59**	**51.37**

Latency and memory

Qwen3-VL-2B (FP8) records and plans; Qwen3-VL-8B answers. Optional 8-frame clips: 8.97 fps.



- Recording: **0.64 s** per second of video.
- Answering: **0.56–0.57 s**, with 9.6 ms of routing.
- Memory: **4.0 MB** per hour (visual KV cache: 118.65 GB).